

 PROJETS

LocalOps AI — Privacy-Preserving Incident Triage & Root Cause Analysis System

Local incident triage and root cause analysis system based on an SLM fine-tuned with QLoRA, integrating severity calibration, RAG on runbooks, FastAPI, Prometheus monitoring, Docker, and a web dashboard.

Development of an intelligent platform for the dynamic classification of prospects and clients using digital twins

Development of an AI platform combining RFM-ES/AHP, CLV, churn, conversion, dynamic classification, personas, A/B testing simulation, PostgreSQL, Streamlit, LangGraph, and a governed local LLM.

AuraMarket — Intelligent Marketplace Platform with Multi-Agent System

Designed and developed an intelligent marketplace platform integrating smart product search, user assistance, automated negotiation, dynamic offers, and security monitoring. The project was based on a multi-agent system where specialized agents collaborate to manage buyer-side negotiation, seller counter-offers, offer generation, and suspicious activity detection. The work included system architecture design, product data processing, intelligent service integration, feature development, and functional testing.

Deep learning-based skin disease detection (CNN)

Automatic skin disease detection system based on deep learning (CNN), capable of analyzing dermatological images to provide a rapid pre-diagnosis with confidence scores and medical recommendations

Virtualization / Cloud Solutions

Design and deployment of an isolated internal network in a virtualized environment using the Proxmox VE Type 1 hypervisor

Bank Transaction Simulator (Java Desktop Application)

Design and implementation of a Java-based desktop application simulating bank transactions, including account management, deposits, withdrawals, and transaction history, with a focus on object-oriented design and concurrency handling

 EXPERIENCE

Data Science & Artificial Intelligence Intern,

Smart Automation Technologies

06/2026 – 08/2026 | tangier

Designed and developed an intelligent platform for dynamic prospect and customer classification based on digital twins. Implemented AHP-weighted RFM-ES scoring, CLV, churn and conversion models, customer segmentation and personas, as well as A/B simulation. Built a PostgreSQL/Streamlit-based architecture with LangGraph multi-agent supervision and a governed local LLM with human validation.

Full-Stack Web Developer,

Smart Automation Technologies

12/2023 – 10/2025 | Tangier, Morocco

Development and maintenance of web applications using full-stack technologies. Design of user interfaces to enhance user experience. Server performance improvement and optimization of backend processes

 EDUCATION

Engineering Program, EMSI Tangier

10/2025 – aujourd'hui | Tangier

Bachelor of Science and Technology (Computer Engineering), Faculty of Sciences and Technology

2022 – 2023 | Tangier, Morocco

 SKILLS

Deep learning — TensorFlow, Keras

Machine Learning — Supervised and Unsupervised Learning

Design and Infographics — Pack Adobe: Photoshop, Premier Pro, Light Room

Programming Languages — Python, Java, TypeScript, C++, SQL

Machine Learning & AI — PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib, NLP, QLoRA, Ollama

Big Data & Databases — Apache Hadoop (HDFS), Apache Spark, Apache Kafka, PostgreSQL

Web & Software Architecture — Spring Boot, Angular, Microservices, Streamlit

Generative AI & LLMs — Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems

 LANGUAGES

English, French

CHELLE ZAKARIAE

zakariaechelle2001@gmail.com